BANGALORE INSTITUTE OF TECHNOLOGY

Department of Computer Science & Engineering

Course: Project Management with GIT (BCSL307A / BCS358C)

ACADEMIC LABORATORY RECORD (EXPERIMENTS 1 TO 7)

Student Name:	Skanda K N
USN:	1BI25CS252
Terminal Identifier:	skandakn1BI25CS252@1BI25CS252-Skanda
GitHub Repository:	https://github.com/skandaknbit/BCS358C-Project-Management-with-Git

Experiment 1. Setting Up and Basic Commands

Aim / Problem Statement:

Initialize a new Git repository in a dedicated directory. Create a new file, add it to the staging area, and commit the changes with an appropriate commit message.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp1
```

- **Step 2: Navigate into the project folder**

```
$ cd exp1
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Configure user credentials globally**

```
$ git config --global user.name "Skanda K N (1BI25CS252)"
$ git config --global user.email "skandakn13@gmail.com"
```

- **Step 5: Create a new sample file**

```
$ echo "Hi, this is Skanda K N (1BI25CS252)" > sample.txt
```

- **Step 6: Display file contents to verify creation**

```
$ cat sample.txt
```

- **Step 7: Add the file to the staging area (Index)**

```
$ git add sample.txt
```

- **Step 8: Commit the staged file with an intention message**

```
$ git commit -m "Initial commit"
```

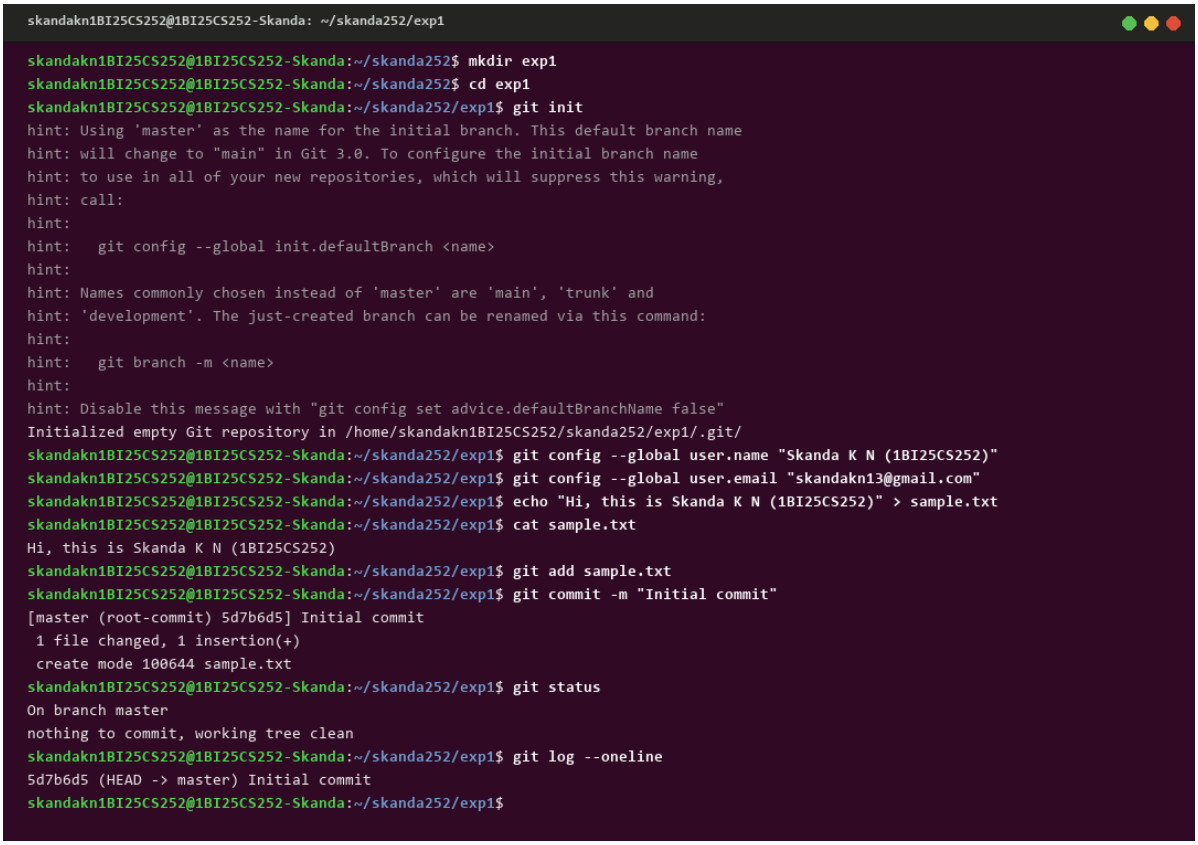
- **Step 9: Verify the working tree status**

```
$ git status
```

- **Step 10: View the chronological commit log in compressed single-line format**

```
$ git log --oneline
```

Execution Terminal Window Screenshot:


skandakn1BI25CS252@1BI25CS252-Skanda: ~/skanda252/exp1
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ mkdir exp1
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ cd exp1
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/skandakn1BI25CS252/skanda252/exp1/.git/
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git config --global user.name "Skanda K N (1BI25CS252)"
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git config --global user.email "skandakn13@gmail.com"
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ echo "Hi, this is Skanda K N (1BI25CS252)" > sample.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ cat sample.txt
Hi, this is Skanda K N (1BI25CS252)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git add sample.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git commit -m "Initial commit"
[master (root-commit) 5d7b6d5] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 sample.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git status
On branch master
nothing to commit, working tree clean
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$ git log --oneline
5d7b6d5 (HEAD -> master) Initial commit
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp1$

Experiment 2. Creating and Managing Branches

Aim / Problem Statement:

Create a new branch named 'feature-branch'. Switch to the 'master/main' branch. Merge the 'feature-branch' into 'master/main'.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp2
```

- **Step 2: Navigate into the directory**

```
$ cd exp2
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Switch to or establish the master branch**

```
$ git checkout -b master
```

- **Step 5: Create a baseline file on master**

```
$ echo "Initial content on master" > mainfile.txt
```

- **Step 6: Stage and commit the baseline file**

```
$ git add mainfile.txt
$ git commit -m "Initial commit on master"
```

- **Step 7: Create a new branch named 'feature-branch'**

```
$ git branch feature-branch
```

- **Step 8: Switch to the newly created feature-branch**

```
$ git checkout feature-branch
```

- **Step 9: Create and write new feature content**

```
$ echo "New feature content" > feature.txt
```

- **Step 10: Stage and commit the feature file on feature-branch**

```
$ git add feature.txt
$ git commit -m "Add feature.txt on feature-branch"
```

- **Step 11: Switch back to the master branch**

```
$ git checkout master
```

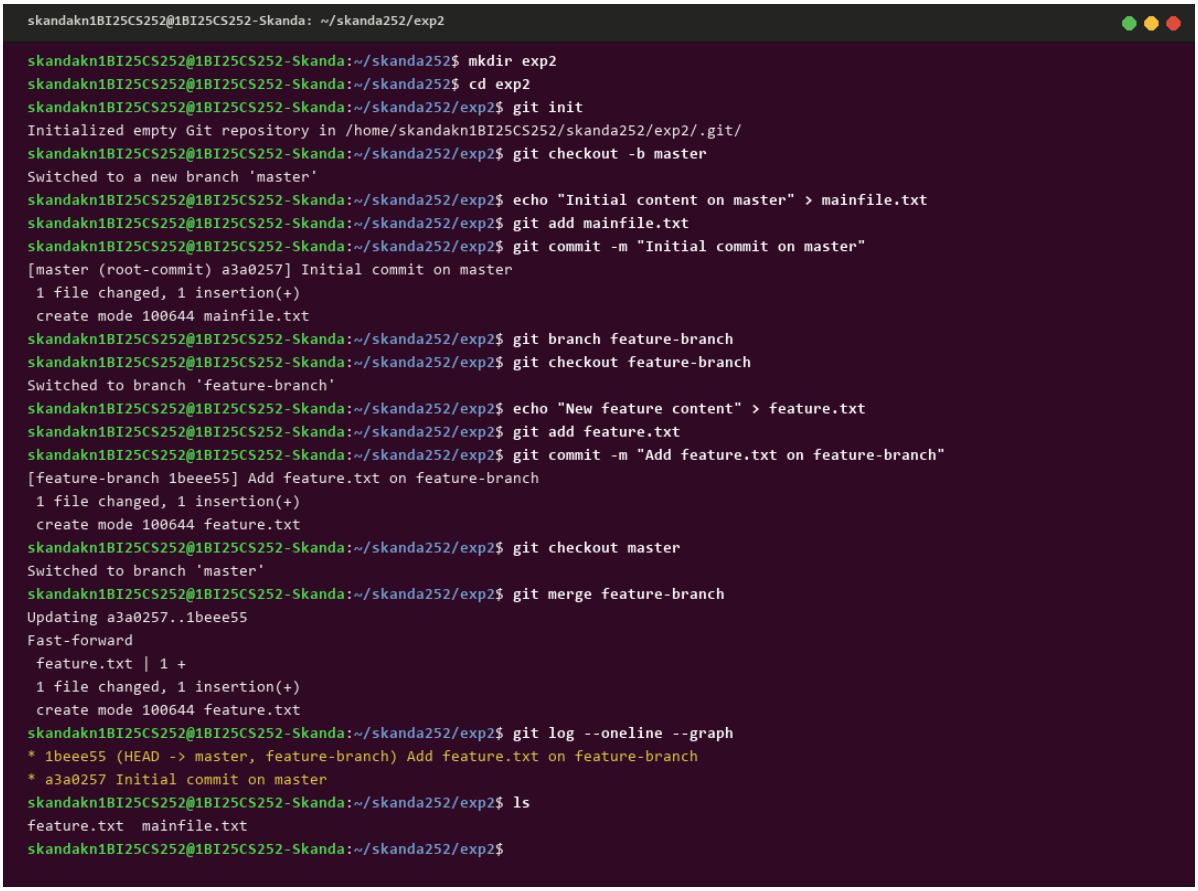
- **Step 12: Merge feature-branch into master**

```
$ git merge feature-branch
```

- **Step 13: View the commit graph and verify files**

```
$ git log --oneline --graph
$ ls
```

Execution Terminal Window Screenshot:



Experiment 3. Stashing Changes

Aim / Problem Statement:

Write the commands to stash uncommitted modifications in the current branch, switch branches, switch back, and re-apply the stashed modifications.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp3
```

- **Step 2: Navigate into the directory**

```
$ cd exp3
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Switch to master and make a base commit**

```
$ git checkout -b master
$ echo "Base code for application" > app.txt
$ git add app.txt
$ git commit -m "Base commit"
```

- **Step 5: Create a secondary branch for hotfixes**

```
$ git branch bugfix-branch
```

- **Step 6: Make uncommitted, experimental changes in the working directory**

```
$ echo "Underconstruction draft code..." >> app.txt
```

- **Step 7: Check working tree status to observe dirty state**

```
$ git status
```

- **Step 8: Stash uncommitted changes onto the stash stack**

```
$ git stash
```

- **Step 9: Switch to the bugfix branch with a clean working tree**

```
$ git checkout bugfix-branch
```

- **Step 10: Switch back to master**

```
$ git checkout master
```

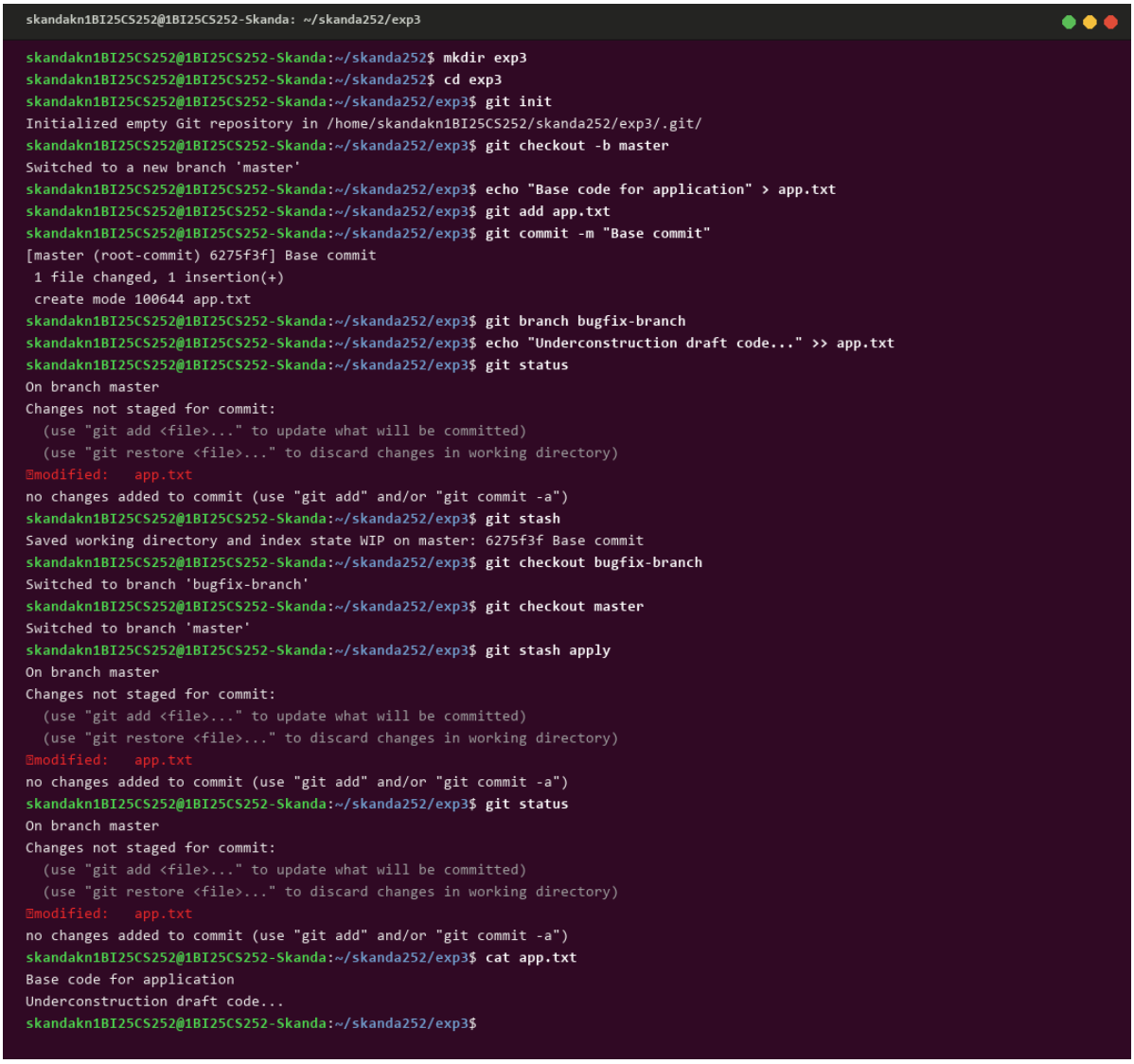
- **Step 11: Re-apply the stashed modifications**

```
$ git stash apply
```

- **Step 12: Verify that modifications are restored**

```
$ git status
$ cat app.txt
```

Execution Terminal Window Screenshot:



Experiment 4. Collaboration and Remote Repositories (Cloning)

Aim / Problem Statement:

Clone an existing remote Git repository from GitHub to your local machine, inspect remote bindings, and examine the historical commit log.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp4
```

- **Step 2: Navigate into the directory**

```
$ cd exp4
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Clone the remote repository from GitHub**

```
$ git clone https://github.com/skandaknbit/BCS358C-Project-Management-with-Git.git
```

- **Step 5: Inspect directory structure of the cloned repository**

```
$ tree BCS358C-Project-Management-with-Git
```

- **Step 6: Enter the cloned project folder**

```
$ cd BCS358C-Project-Management-with-Git
```

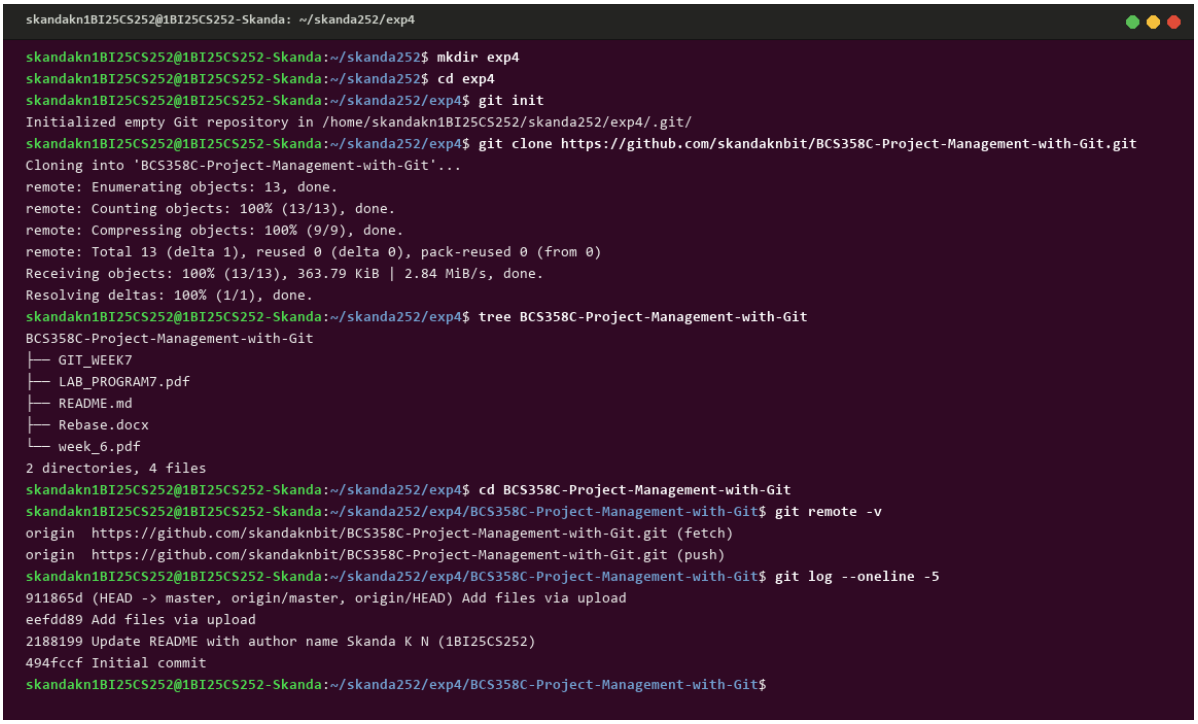
- **Step 7: Verify remote repository connection and URLs**

```
$ git remote -v
```

- **Step 8: View the last 5 commits from the cloned history**

```
$ git log --oneline -5
```

Execution Terminal Window Screenshot:


skandakn1BI25CS252@1BI25CS252-Skanda: ~/skanda252/exp4
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ mkdir exp4
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ cd exp4
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4$ git init
Initialized empty Git repository in /home/skandakn1BI25CS252/skanda252/exp4/.git/
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4$ git clone https://github.com/skandaknbit/BCS358C-Project-Management-with-Git.git
Cloning into 'BCS358C-Project-Management-with-Git'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), 363.79 KiB | 2.84 MiB/s, done.
Resolving deltas: 100% (1/1), done.
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4$ tree BCS358C-Project-Management-with-Git
BCS358C-Project-Management-with-Git
├── GIT_WEEK7
├── LAB_PROGRAM7.pdf
├── README.md
├── Rebase.docx
└── week_6.pdf
2 directories, 4 files
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4$ cd BCS358C-Project-Management-with-Git
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4/BCS358C-Project-Management-with-Git$ git remote -v
origin https://github.com/skandaknbit/BCS358C-Project-Management-with-Git.git (fetch)
origin https://github.com/skandaknbit/BCS358C-Project-Management-with-Git.git (push)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4/BCS358C-Project-Management-with-Git$ git log --oneline -5
911865d (HEAD -> master, origin/master, origin/HEAD) Add files via upload
eefdd89 Add files via upload
2188199 Update README with author name Skanda K N (1BI25CS252)
494fccf Initial commit
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp4/BCS358C-Project-Management-with-Git$

Experiment 5. Fetching and Rebasing

Aim / Problem Statement:

Fetch the latest changes from the remote repository and rebase your local branch onto the updated base branch to maintain a clean, linear commit history.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp5
```

- **Step 2: Navigate into the directory**

```
$ cd exp5
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Create root commit on baseline branch**

```
$ echo "1st commit" > README.md
$ git add .
$ git commit -m "1st"
```

- **Step 5: Create and checkout feature branch**

```
$ git checkout -b feature
$ echo "branch commit" > README.md
$ git add .
$ git commit -m "2nd"
```

- **Step 6: Switch to master and make an independent commit**

```
$ git checkout -b master
$ echo "Master commit" > master.txt
$ git add .
$ git commit -m "3rd"
```

- **Step 7: Switch back to feature branch**

```
$ git checkout feature
```

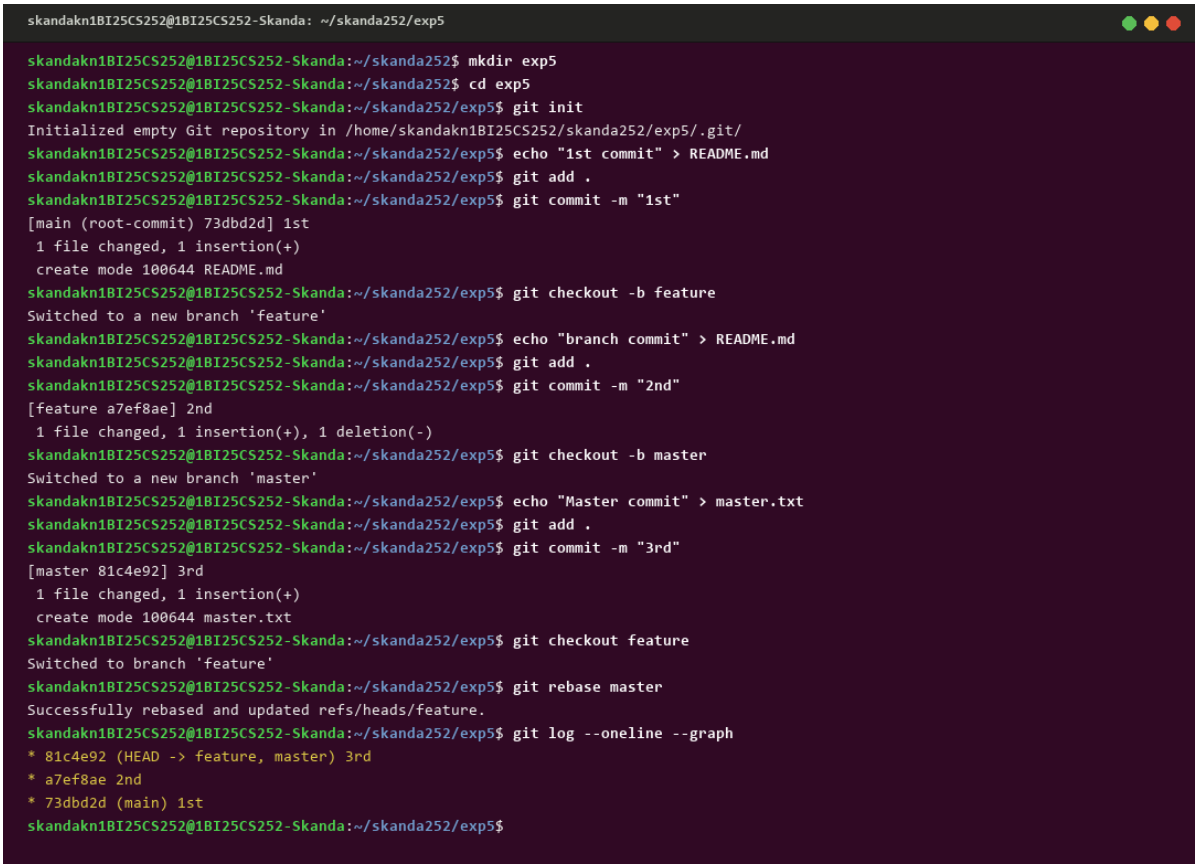
- **Step 8: Rebase feature branch on top of master**

```
$ git rebase master
```

- **Step 9: Inspect the linear commit graph**

```
$ git log --oneline --graph
```

Execution Terminal Window Screenshot:


skandakn1BI25CS252@1BI25CS252-Skanda: ~/skanda252/exp5
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ mkdir exp5
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ cd exp5
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git init
Initialized empty Git repository in /home/skandakn1BI25CS252/skanda252/exp5/.git/
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ echo "1st commit" > README.md
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git add .
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git commit -m "1st"
[main (root-commit) 73dbd2d] 1st
1 file changed, 1 insertion(+)
create mode 100644 README.md
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git checkout -b feature
Switched to a new branch 'feature'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ echo "branch commit" > README.md
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git add .
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git commit -m "2nd"
[feature a7ef8ae] 2nd
1 file changed, 1 insertion(+), 1 deletion(-)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git checkout -b master
Switched to a new branch 'master'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ echo "Master commit" > master.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git add .
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git commit -m "3rd"
[master 81c4e92] 3rd
1 file changed, 1 insertion(+)
create mode 100644 master.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git checkout feature
Switched to branch 'feature'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git rebase master
Successfully rebased and updated refs/heads/feature.
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$ git log --oneline --graph
* 81c4e92 (HEAD -> feature, master) 3rd
* a7ef8ae 2nd
* 73dbd2d (main) 1st
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp5$

Experiment 6. Collaboration and Merging with Custom Commit Messages

Aim / Problem Statement:

Write the command to merge 'feature-branch' into 'master' while providing an informative custom commit message for the merge.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp6
```

- **Step 2: Navigate into the directory**

```
$ cd exp6
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Switch to master and create a project baseline**

```
$ git checkout -b master
$ echo "Project baseline structure" > index.html
$ git add index.html
$ git commit -m "Initial commit"
```

- **Step 5: Create and checkout feature-branch**

```
$ git checkout -b feature-branch
```

- **Step 6: Develop authentication module on feature branch**

```
$ echo "Adding user authentication module" > auth.js
$ git add auth.js
$ git commit -m "Add authentication module"
```

- **Step 7: Switch back to master branch**

```
$ git checkout master
```

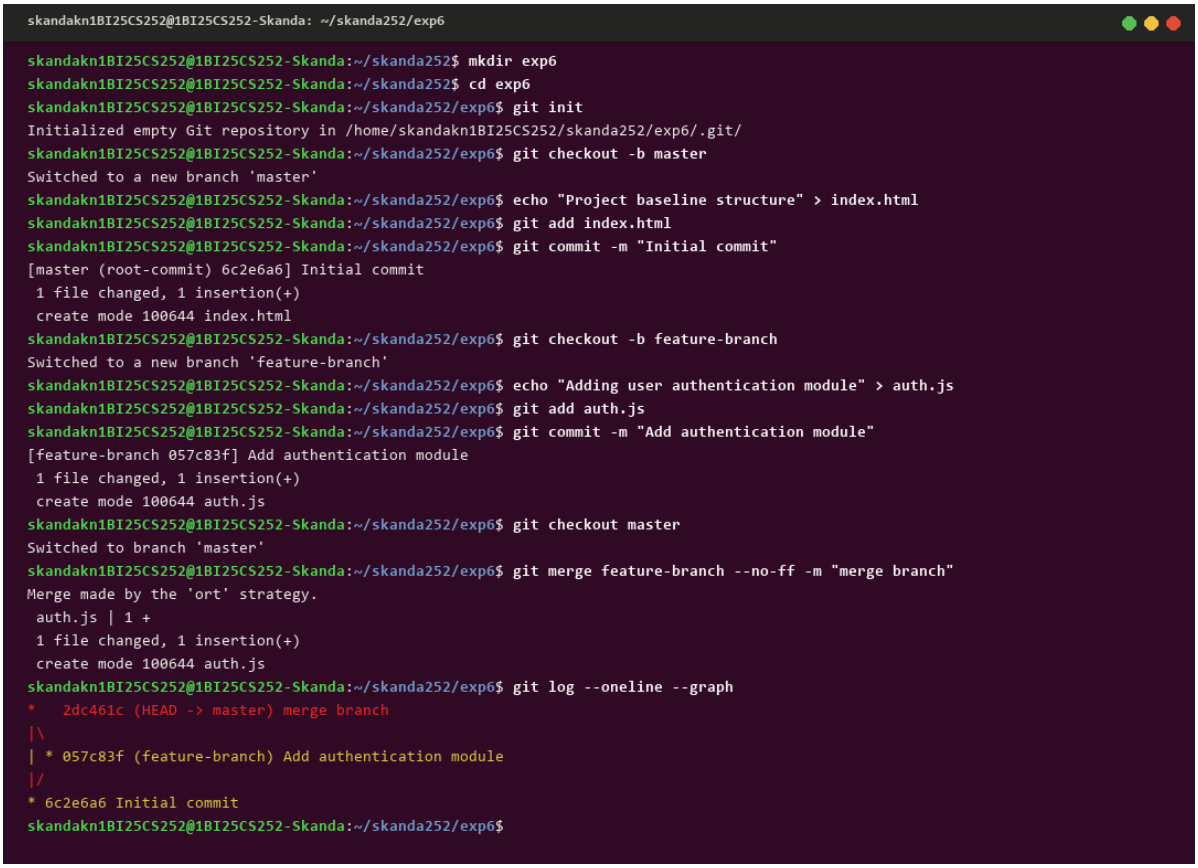
- **Step 8: Perform non-fast-forward merge with a custom message**

```
$ git merge feature-branch --no-ff -m "merge branch"
```

- **Step 9: View the commit graph showing the explicit merge node**

```
$ git log --oneline --graph
```

Execution Terminal Window Screenshot:


skandakn1BI25CS252@1BI25CS252-Skanda: ~/skanda252/exp6
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ mkdir exp6
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ cd exp6
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git init
Initialized empty Git repository in /home/skandakn1BI25CS252/skanda252/exp6/.git/
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git checkout -b master
Switched to a new branch 'master'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ echo "Project baseline structure" > index.html
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git add index.html
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git commit -m "Initial commit"
[master (root-commit) 6c2e6a6] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 index.html
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ echo "Adding user authentication module" > auth.js
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git add auth.js
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git commit -m "Add authentication module"
[feature-branch 057c83f] Add authentication module
1 file changed, 1 insertion(+)
create mode 100644 auth.js
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git checkout master
Switched to branch 'master'
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git merge feature-branch --no-ff -m "merge branch"
Merge made by the 'ort' strategy.
auth.js | 1 +
1 file changed, 1 insertion(+)
create mode 100644 auth.js
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$ git log --oneline --graph
* 2dc461c (HEAD -> master) merge branch
|\
| * 057c83f (feature-branch) Add authentication module
|/
* 6c2e6a6 Initial commit
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp6$

Experiment 7. Git Tags and Releases

Aim / Problem Statement:

Write the command to create a lightweight Git tag named 'v1.0' for a commit in your local repository.

Step-by-Step Execution Procedure:

- **Step 1: Create a separate directory for the experiment**

```
$ mkdir exp7
```

- **Step 2: Navigate into the directory**

```
$ cd exp7
```

- **Step 3: Initialize the Git repository**

```
$ git init
```

- **Step 4: Configure user credentials**

```
$ git config --global user.name "Skanda K N (1BI25CS252)"
$ git config --global user.email "skandakn13@gmail.com"
```

- **Step 5: Verify configuration settings**

```
$ git config --list
```

- **Step 6: Create the first file**

```
$ echo "Hello, I'm Skanda K N (1BI25CS252)" >> file1.txt
$ cat file1.txt
```

- **Step 7: Add and commit the first file**

```
$ git add file1.txt
$ git commit -m "1st commit"
```

- **Step 8: Create the second file**

```
$ echo "Skanda is from BIT (1BI25CS252)" >> file2.txt
$ cat file2.txt
```

- **Step 9: Add and commit the second file**

```
$ git add file2.txt
$ git commit -m "2nd commit"
```

- **Step 10: View the chronological commit log**

```
$ git log --oneline
```

- **Step 11: Create a lightweight Git tag**

```
$ git tag v1.0
```

- **Step 12: Verify the tag creation**

```
$ git tag
```

- **Step 13: View the details of the tagged commit**

```
$ git show v1.0
```

- **Step 14: Connect the local repository to GitHub**

```
$ git remote add origin https://github.com/skandaknbit/trial.git
```

- **Step 15: Push commits and release tags to GitHub**

```
$ git branch -M main
$ git push -u origin main
$ git push origin v1.0
```

Execution Terminal Window Screenshot:

```
skandakn1BI25CS252@1BI25CS252-Skanda: ~/skanda252/exp7
```

```
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ mkdir exp7
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252$ cd exp7
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git init
Initialized empty Git repository in /home/skandakn1BI25CS252/skanda252/exp7/.git/
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git config --global user.name "Skanda K N (1BI25CS252)"
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git config --global user.email "skandakn13@gmail.com"
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git config --list
user.name=Skanda K N (1BI25CS252)
user.email=skandakn13@gmail.com
advice.defaultbranchname=false
init.defaultbranch=main
core.filemode=true
core.bare=false
core.logallrefupdates=true
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ echo "Hello, I'm Skanda K N (1BI25CS252)" >> file1.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ cat file1.txt
Hello, I'm Skanda K N (1BI25CS252)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git add file1.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git commit -m "1st commit"
[main (root-commit) dd3fb60] 1st commit
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ echo "Skanda is from BIT (1BI25CS252)" >> file2.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ cat file2.txt
Skanda is from BIT (1BI25CS252)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git add file2.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git commit -m "2nd commit"
[main 6509252] 2nd commit
 1 file changed, 1 insertion(+)
 create mode 100644 file2.txt
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git log --oneline
6509252 (HEAD -> main) 2nd commit
dd3fb60 1st commit
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git tag v1.0
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git tag
v1.0
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$ git show v1.0
commit 6509252a4702e52534abee58ccaa28eeccf35a89 (HEAD -> main, tag: v1.0)
Author: Skanda K N (1BI25CS252) <skandakn13@gmail.com>
Date:   Tue Nov 10 13:26:15 2026 +0530

    2nd commit

diff --git a/file2.txt b/file2.txt
new file mode 100644
index 0000000..23b0826
--- /dev/null
+++ b/file2.txt
@@ -0,0 +1 @@
+Skanda is from BIT (1BI25CS252)
skandakn1BI25CS252@1BI25CS252-Skanda:~/skanda252/exp7$
```